A Quantum Algorithm for the Simple Harmonic Oscillator

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Abstract

We implement a quantum algorithm to solve the Simple Harmonic Oscillator (SHO) using the **Linear Combination of Unitaries (LCU)** framework from Xin et al. (2020). The second-order ODE $y'' + \omega^2 y = 0$, y(0)=1, y'(0)=1, ω=1 is reduced to a first-order matrix system and the matrix exponential e^{At} is approximated via a Taylor expansion. Encoded on a **4-qubit Classiq circuit**, the quantum solution closely tracks the classical trajectory with energy conservation confirmed throughout t ∈ [0,1]. A systematic parameter study identifies **k=5** Taylor terms, **bound=0.01**, and **n_{shots}=8192** as the optimal operating point.

Problem & Quantum Approach

ODE System

$y'' + \omega^2 y = 0,\ \ \omega=1,\ \ y(0)=1,\ \ y'(0)=1$

Matrix Form

dx/dt = Ax, $x(0) = [1, 1]^T$, A = [[0,1],[-1,0]]

Solution: $x(t) = e^{At}x(0)$

Taylor Approximation (k=5)

$e^{At} \approx \Sigma(m=0..k)\ (At)^m/m!$

Coefficients decay to <0.01% by m=5 — k=5 is sufficient for t ∈ [0,1].

LCU Circuit (3 Steps)

1. PREPARE: Load normalised Taylor coefficients as quantum superposition on the controller register via inplace_prepare_state()

2. SELECT: Apply A^m to work register, controlled on |m⟩

3. PREPARE†: Uncompute controller (handled automatically by within_apply)

Post-Selection

Keep only shots where the controller returns to |0⟩. Recover y(t) and v(t) from measurement probabilities using classical sign correction.

Note: A is unitary (A†A=I), placing this in Case I of Xin et al. — the most efficient LCU case.

Results & Energy Analysis

At k=5, bound=0.01, n_{shots}=8192, the quantum solution closely matches the classical trajectory for both position y(t) and velocity v(t) across t ∈ [0,1]. Total energy $E(t) = E_k + E_p = ½v^2 + ½\omega^2 y^2$ remains approximately flat, confirming **energy conservation**. Post-selection success rate decreases from ~87% at t=0 to ~13% at t=1, consistent with the theoretical lower bound $1/\Pi^2$ where Π = Σ Taylor coefficients.

Parameter Study Summary

Parameter	Range Tested	Optimal	Key Finding
bound	0.001 – 0.1	**0.01**	No measurable effect on accuracy. The Taylor coefficient distribution at k=5 is sparse enough that Classiq compiles essentially the same circuit regardless of the bound.
Taylor order k	1 – 8	**5**	Dominant accuracy control. Width grows O(log k), depth O(k log k). k=5 sits at the accuracy/depth sweet spot before the expensive depth jump at k=8.
n_{shots}	256 – 8192	**8192**	Shot noise is not the bottleneck at k=5 — Taylor truncation dominates error. All trajectories look nearly identical across shot counts. 8192 is a safe default.

Circuit Resources at Optimal Parameters (k=5)

Metric	Value	Metric	Value
Circuit depth	~260 gates	Post-selection rate at t=0.5	~37%
Circuit width	4 qubits	Post-selection rate at t=1.0	~13%

Real-World Applications & Scaling

The same LCU algorithm applies to any first-order linear system dx/dt = Ax, scaling as O(poly(log N)) vs. classical $O(N^3)$, an exponential speedup for large N. **Protein dynamics** (Liu et al., 2024): molecular stiffness matrix with $N{\sim}10^3$–10^4, directly applicable to drug discovery and misfolding diseases. **Heat conduction** (Xin et al., 2023): spatial Laplacian on a 2D grid; Xin himself validated the LCU approach experimentally on a nuclear spin processor. **Hardware cost:** ~$7.67 (AWS Braket Rigetti) to ~$246 (IonQ Aria) per 8192-shot job today for our own SHO to be run on a modern quantum computer..

Reference: Xin et al. (2020), Phys. Rev. A 101, 032307 | Liu et al. (2024), arXiv:2411.03972 | Xin et al. (2023), Science Bulletin